6 **(a)** A beam of white light, consisting of wavelengths between 400 nm and 700 nm, is incident on a cool sodium gas at low pressure. Fig. 6.1 shows the four lowest energy levels of the sodium atom.

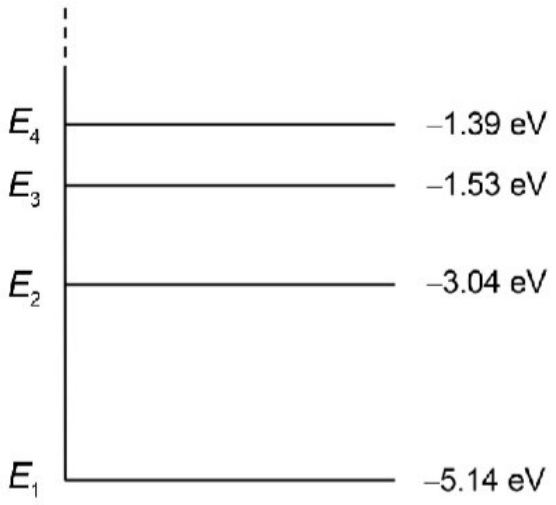


Fig. 6.1

(i) Explain the formation of an absorption line spectrum.

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..... [3]

(ii) Show that the wavelength of the photon that causes the transition from E_1 to E_2 is 592 nm.

[1]

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(iii) Determine the number of absorption spectral lines that are observed. Show your working.

number of observable lines = ... [2]

(b) An electron microscope uses an array of electromagnetic lenses to accelerate and focus a beam of electrons to create an enlarged image of a specimen.

(i) Determine the kinetic energy, in eV, of an electron which has a de Broglie wavelength that is the same as the wavelength of the photon in **(a)(ii)**.

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kinetic energy = ... eV [2]

(ii) With reference to the kinetic energy of an electron, explain one advantage of an electron microscope over an optical microscope.

.....

.....

..... [2]

(c) The radial wavefunction of the outermost electron in a sodium atom is approximated by

$$\psi(r) = Ae^{-r/a},$$

where A and a are constants, and r is the distance from the nucleus.

(i) State the physical meaning of the quantity $|\psi(r)|^2$.

.....

..... [1]

(ii) State and explain where the electron is most likely to be found in the atom.

.....

.....

.....

..... [2]

[Total :13]